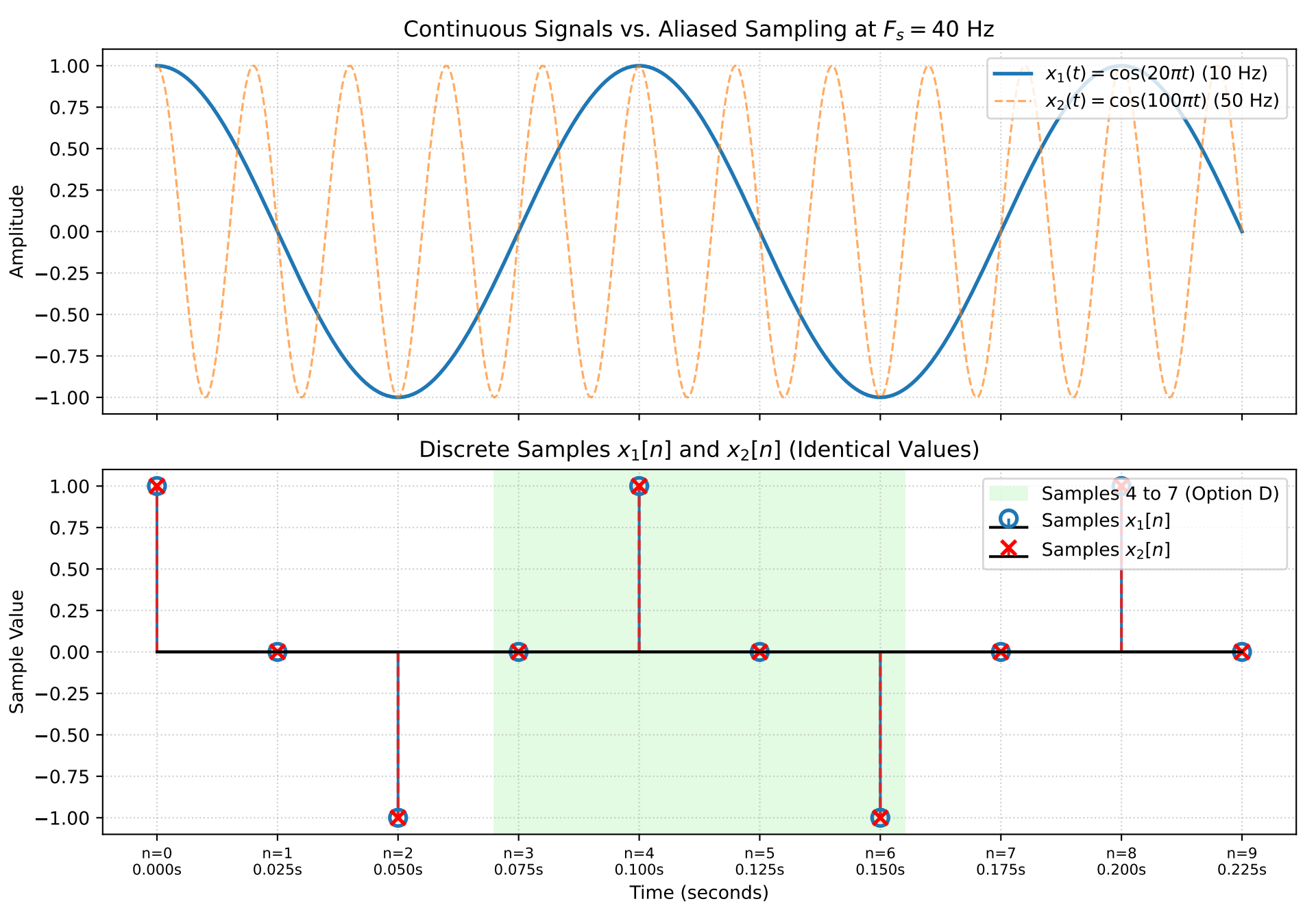
Continuous Signals vs. Aliased Sampling at $F_s = 40$ Hz
Amplitude
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
$x_1(t) = \cos(20\pi t)$ (10 Hz)
$x_2(t) = \cos(100\pi t)$ (50 Hz)
Discrete Samples $x_1[n]$ and $x_2[n]$ (Identical Values)
Sample Value
Samples 4 to 7 (Option D)
Samples $x_1[n]$
Samples $x_2[n]$
n=0
0.000s
n=1
0.025s
n=2
0.050s
n=3
0.075s
n=4
0.100s
n=5
0.125s
n=6
0.150s
n=7
0.175s
n=8
0.200s
n=9
0.225s
Time (seconds)